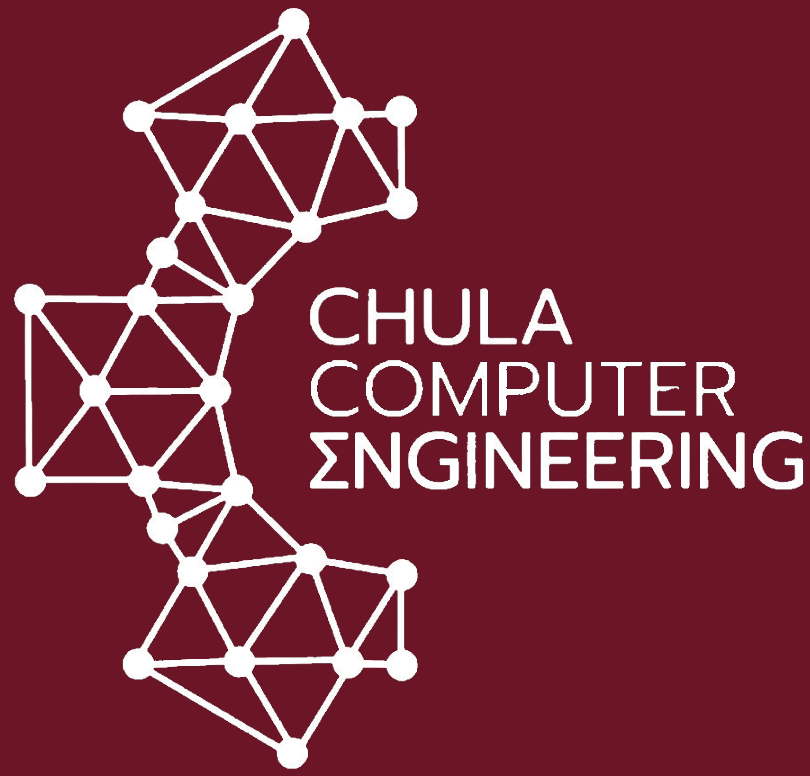


CCES 2026

Chula Computer Engineering Showcase

Friday, 15 May 2026 | Sala Phra Kieo



BUSINESS DOMAIN Finance & Digital Economy

MTL Agency Chatbot: An Intelligent Multi-Agent System for Insurance Sales Support

ระบบแชทบอทสนับสนุนการขายสำหรับตัวแทนเมืองไทยประกันชีวิต

ABSTRACT

MTL Agency Chatbot is a multi-agent conversational AI system built for Muang Thai Life Assurance (MTL) to eliminate the information-access gap faced by sales agents. Under the current workflow, agents search fragmented PDF brochures and LINE messages, causing delays and inaccurate recommendations during customer interactions.

The system implements five specialized agents via LangGraph: a two-layer Guard Agent (regex + LLM) for safety enforcement; a Router Agent for intent classification; a General Agent for FAQs; a Promotion Agent using RAG with hybrid search (70% semantic + 30% BM25) and page-level citation; and a Recommend Agent with a deterministic four-dimension scoring system (Need Match 40%, Age Fit 25%, Occupation Fit 20%, Coverage Breadth 15%) for fully auditable product recommendations. The backend uses FastAPI, PostgreSQL with pgvector, and Gemini 3 Flash — selected via statistical comparison of five LLMs.

Evaluation shows Guard F1 94.8%, Promotion Hit Rate@5 and Citation Accuracy 100%, Recommend MRR 0.957, General CSAT 4.97/5.0, and 0% wrong answer rate end-to-end at 0.09 ฿ per query.

KEY FEATURES

01

Multi-Agent Pipeline

Five specialized agents (Guard → Router → General / Recommend / Promotion) orchestrated via LangGraph with safety-first routing and per-agent evaluation.

02

Explainable Recommendation

Deterministic four-dimension scoring (Need 40%, Age 25%, Occupation 20%, Coverage 15%) — LLM restricted to presenter role. Fully auditable, reproducible.

03

RAG with Citation

Hybrid search (70% semantic + 30% BM25) over promotional PDFs with page-level citation and Gemini Vision OCR for Thai document ingestion.

KEY RESULTS

94.8%

Guard Agent
F1-Score

100%

Citation Accuracy
(0 hallucination)

0.957

Recommend
MRR

0%

Wrong Answer Rate
(End-to-End)

4.97/5

General Agent
CSAT

0.09 ฿

Avg Cost
per Query

SAMPLE OUTPUTS

Admin Portal

Knowledge base upload & publish

Chat Interface

Web chat UI with Nong Sod Sai

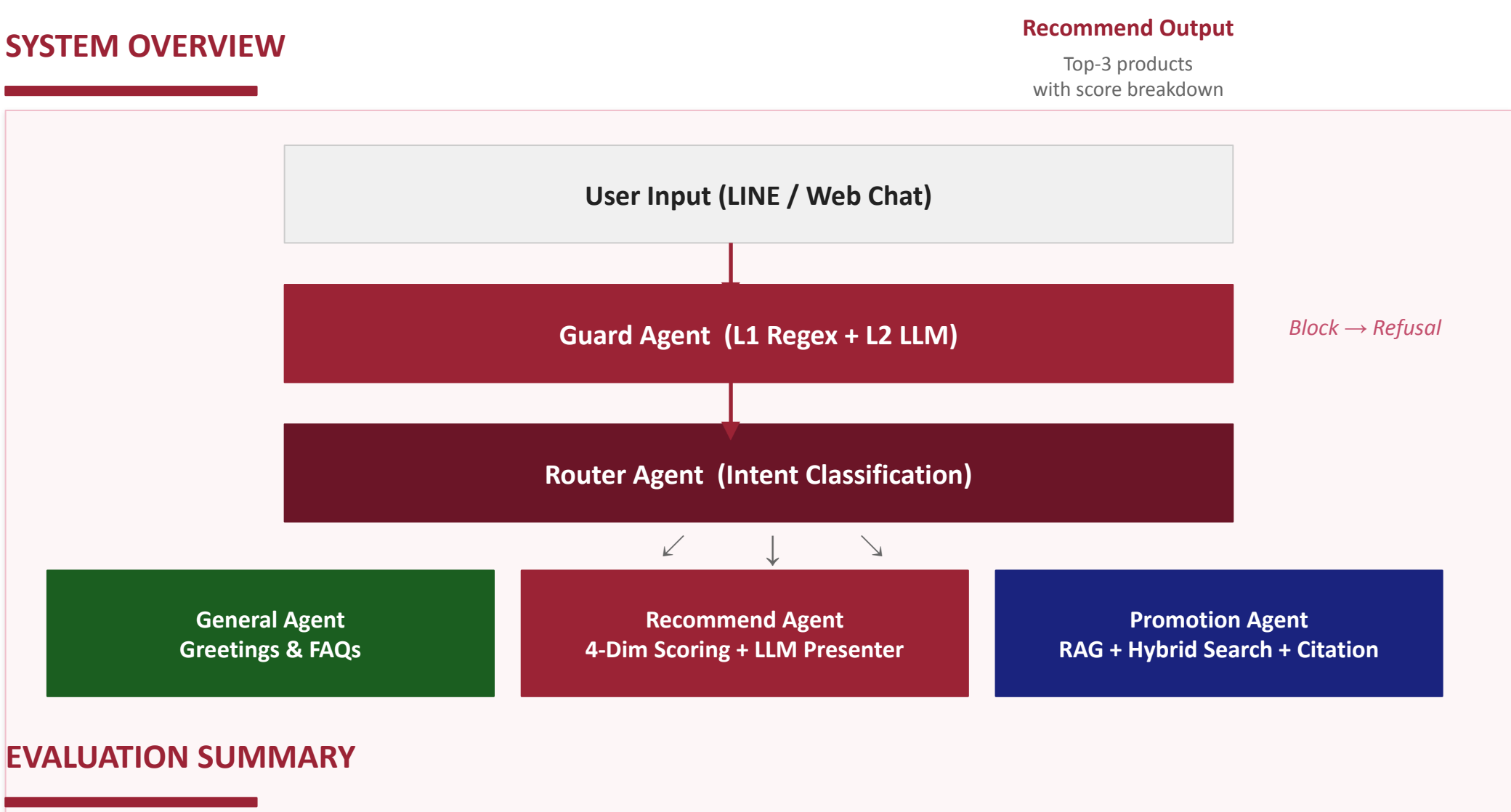
User Management

Manage user roles and chatbot access permissions

Cost & Usage Dashboard

Monitor AI model usage, costs, and agent performance.

SYSTEM OVERVIEW



EVALUATION SUMMARY

Agent	Key Metric	Score	Agent	Key Metric	Score
Guard	F1-Score	94.8%	Recommend	MRR	0.957
Router	Overall Accuracy	92.2%	General	CSAT	4.97 / 5.0
Promotion	Hit Rate@5	100%	End-to-End	Correct Route	90%
Promotion	Citation Accuracy	100%	End-to-End	Avg Cost/Query	0.10 ฿

TEAM MEMBERS

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TECHNOLOGIES

Python, FastAPI, LangGraph,
Next.js, PostgreSQL + pgvector,
Gemini 3 Flash, OpenRouter,
LINE API, Docker

KNOWLEDGE AREA

G1.1 Artificial
Intelligence (AI)